| Time: 1 hr.<br>Paper Code: CSC-501 | B. Tech. (V Semester) I[st] Sessional Test 2026-27<br>Object Oriented Programming | Max. Marks: 12 |
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**Note: Attempt All Questions. Make suitable assumptions wherever necessary.**

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| Q1. CO1 | (a) What is the Object-Oriented Programming paradigm? Explain its key features with suitable examples.<br>(b) Define the term Polymorphism. Differentiate between Compile-Time Polymorphism and Run-Time Polymorphism. | [2+ BL- |
| Q2. CO2 | Design a C++ program using a class *BedAllocation* to manage hospital bed billing. Use constructor overloading to create patient records based on ward type-General Ward (₹500/day), Semi-Private (₹1500/day), Private Room (₹3000/day), or ICU (₹8000/day + ₹2000 fixed) - with charges calculated automatically. Provide a display() function to show patient details, and use an array of objects to store and display records for four patients admitted to the four different ward types. | [4 BL- |
| Q3. CO3 | Implement a class *Product* that contains a private data member *price* and a constructor to initialize the price of a product. In addition, write the necessary code such that the following driver function runs correctly and performs the desired operations.<br><br>`int main()`<br>`{`<br>`  Product p1(1000), p2(500), p4(1000);`<br><br>`  Product p3 = p1 + p2;`<br><br>`  cout << "Sum of prices: ";`<br>`  p3.display();`<br><br>`  if (p1 > p2)`<br>`  cout << "Product 1 is costlier" << endl;`<br><br>`  if (p1 == p4)`<br>`  cout << "Product 1 and Product 4 have the same price" << endl;`<br><br>`  return 0;`<br>`}` | [4 BL- |